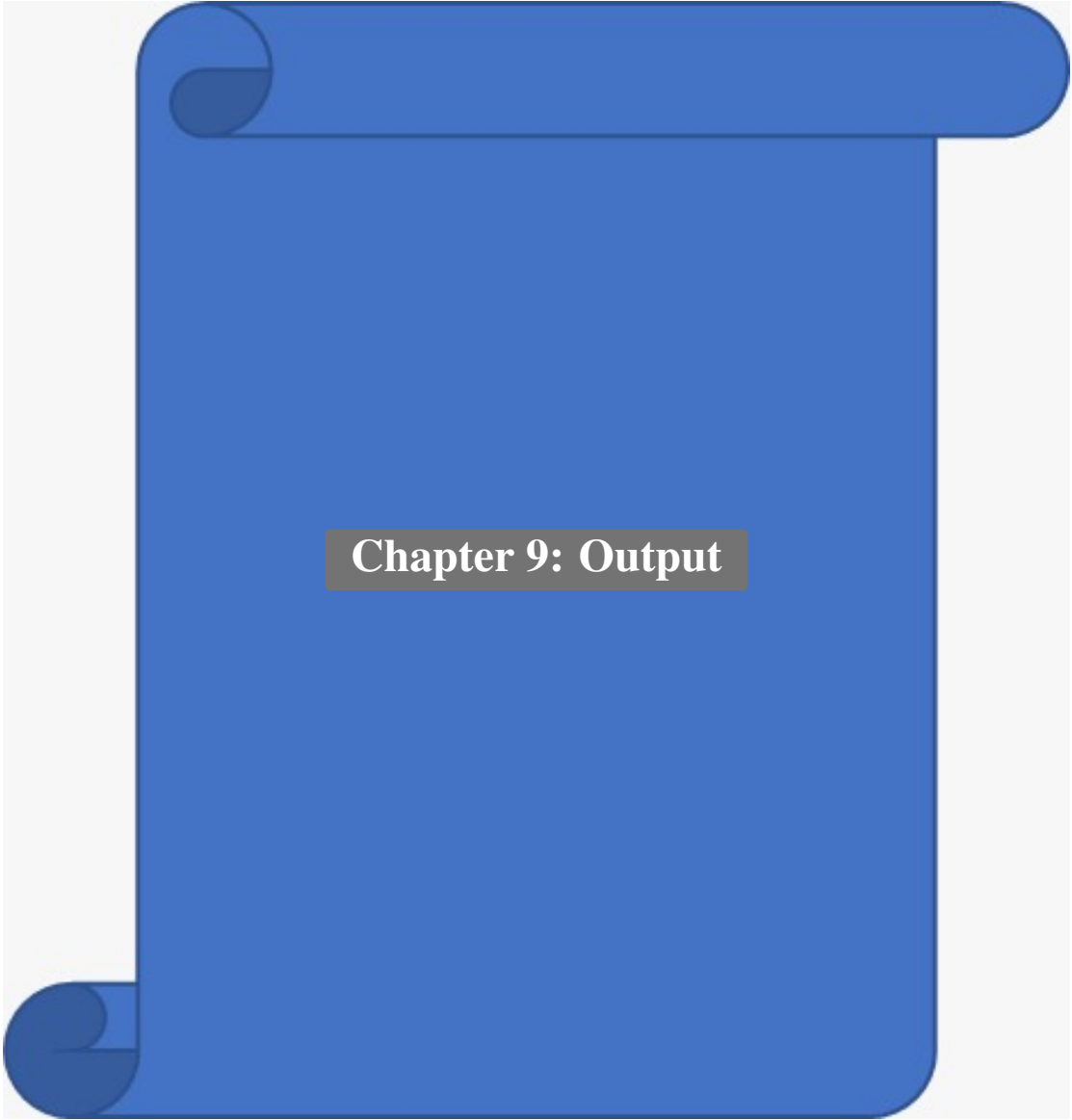




Chapter 9: Output

This chapter presents the output of the AgriTrace platform, describing the user-facing interfaces and features across all modules. The descriptions are organized by functional area and cover the visual design, interactive elements, and data presentation for each screen.

9.1 Project Screenshots

The following screenshots show the key interfaces of the AgriTrace web application, based on the latest provided captures.



Figure 9.1: Landing Page Hero Section

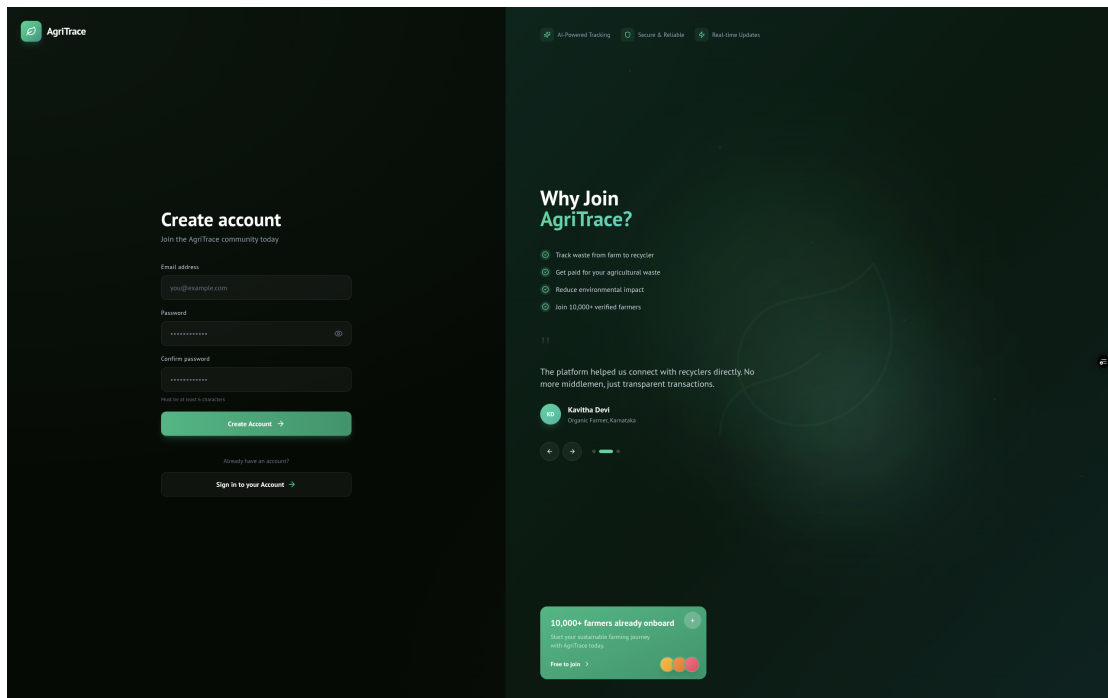


Figure 9.2: Login Page

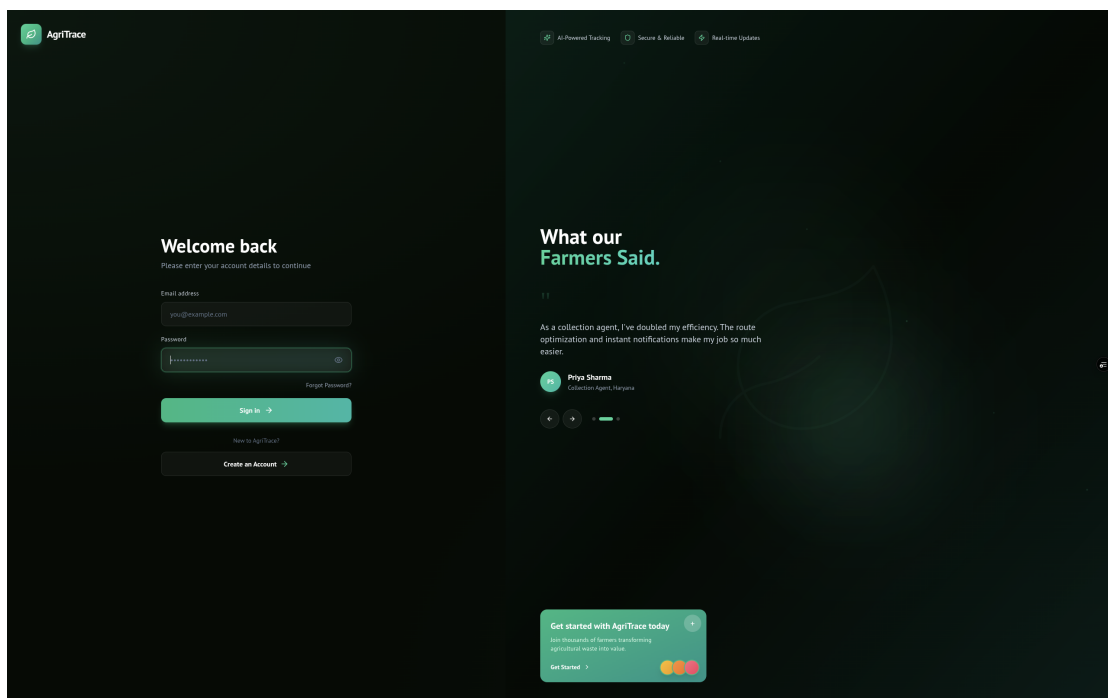


Figure 9.3: Signup Page

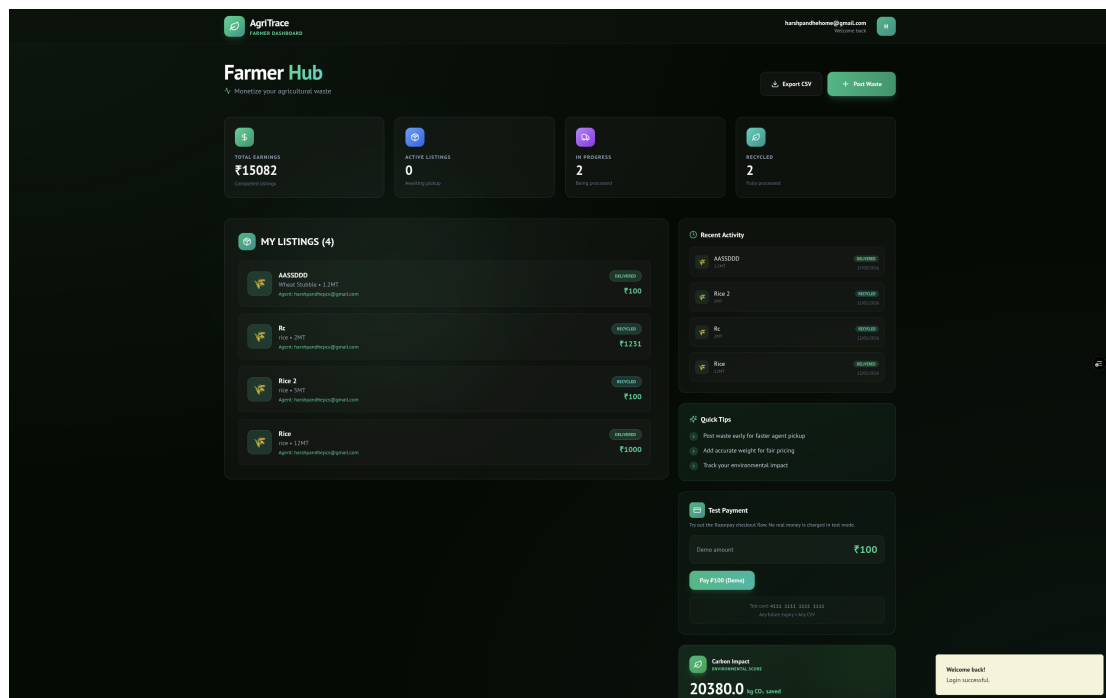


Figure 9.4: Farmer Dashboard

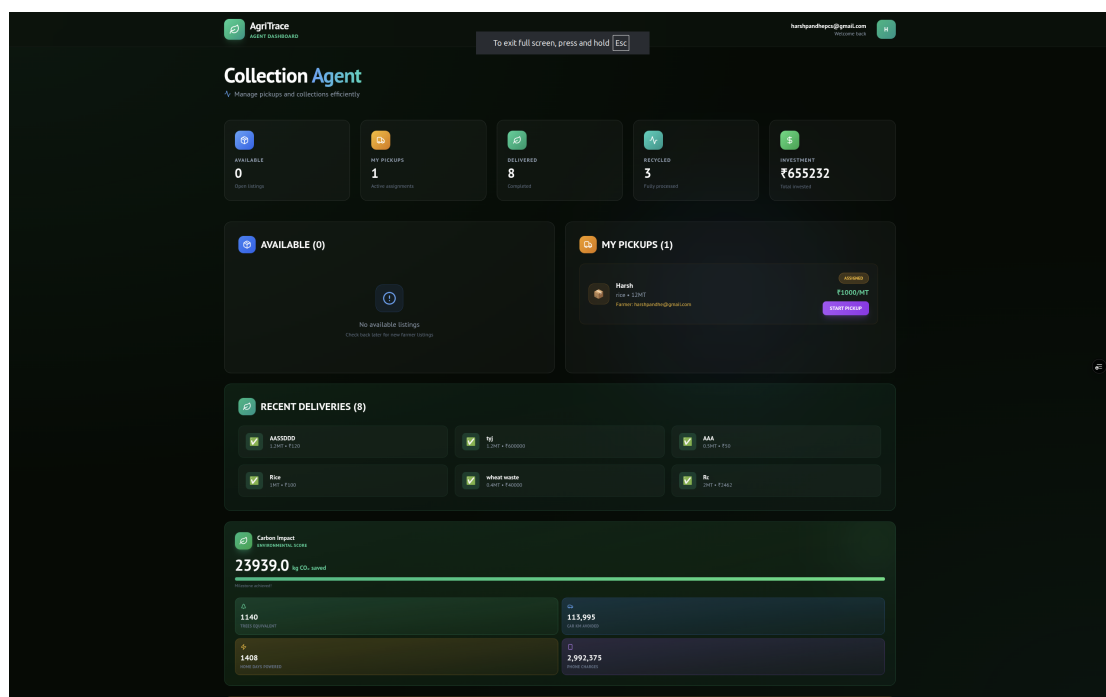


Figure 9.5: Agent Dashboard

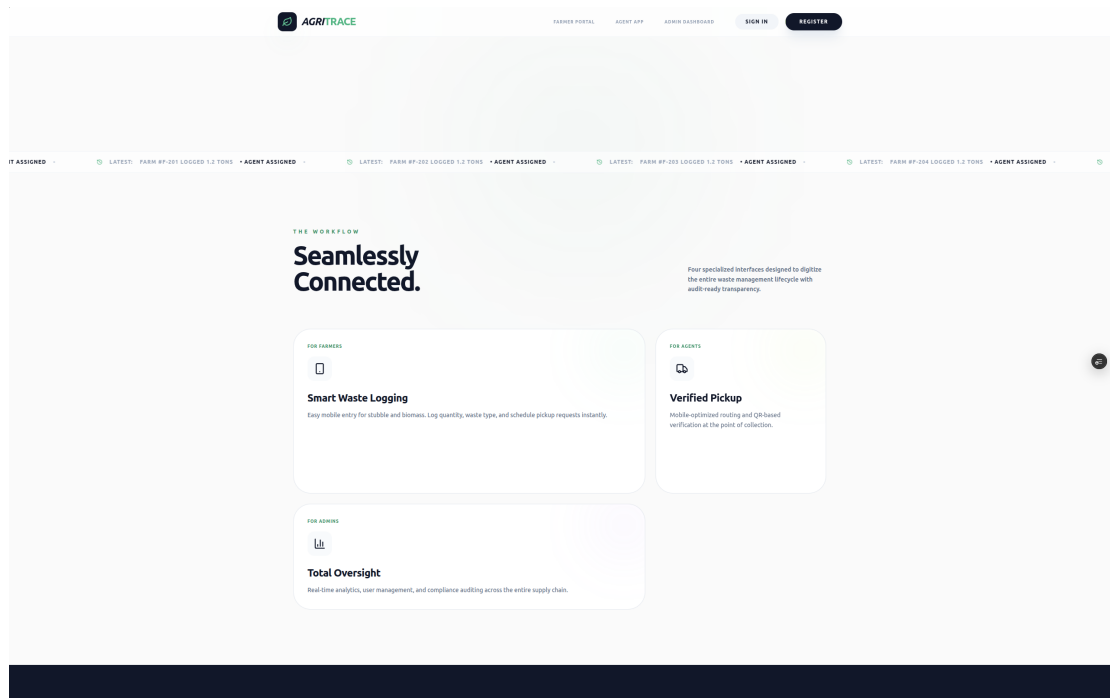


Figure 9.6: Landing Page Workflow Section

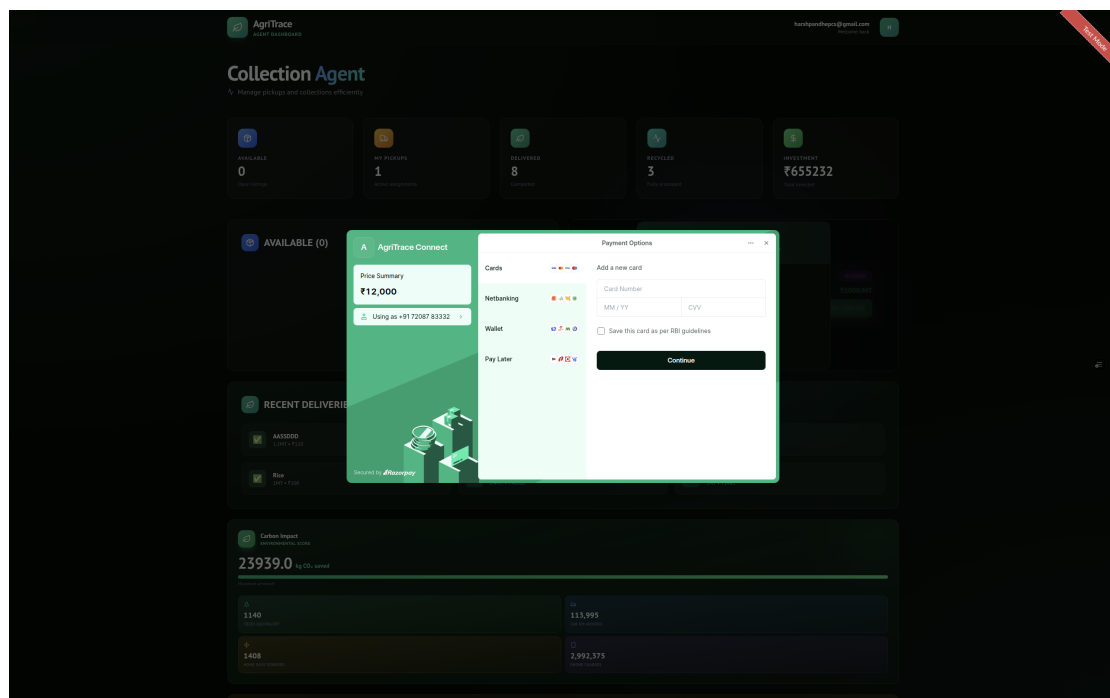


Figure 9.7: Razorpay Payment Gateway Page

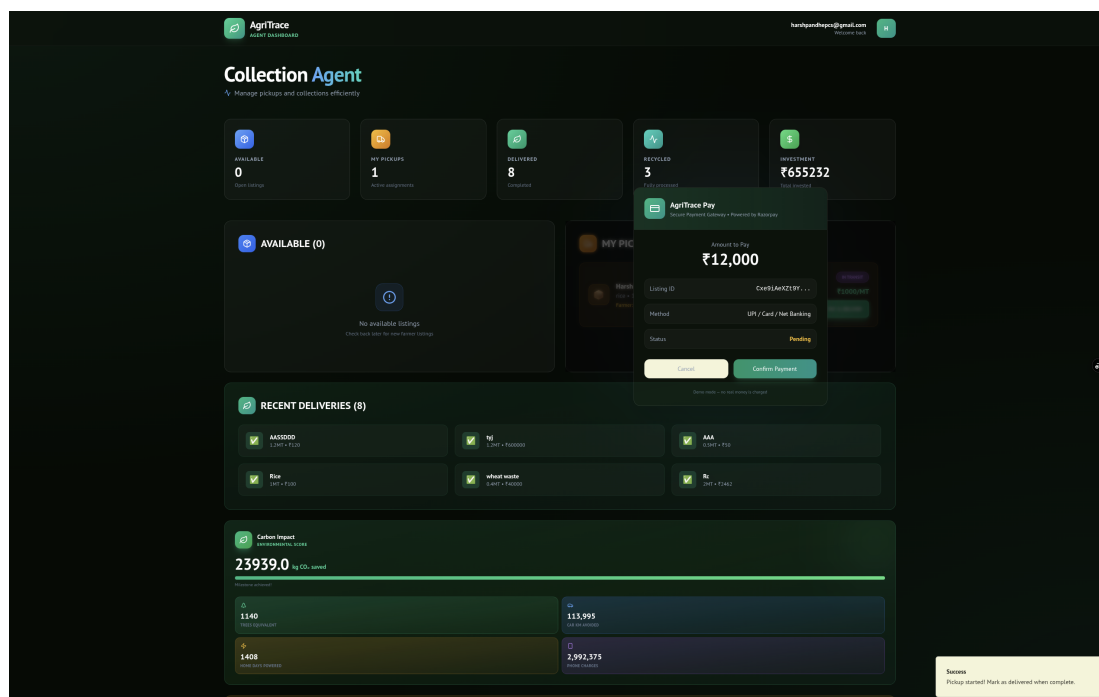


Figure 9.8: Payment Confirmation Page

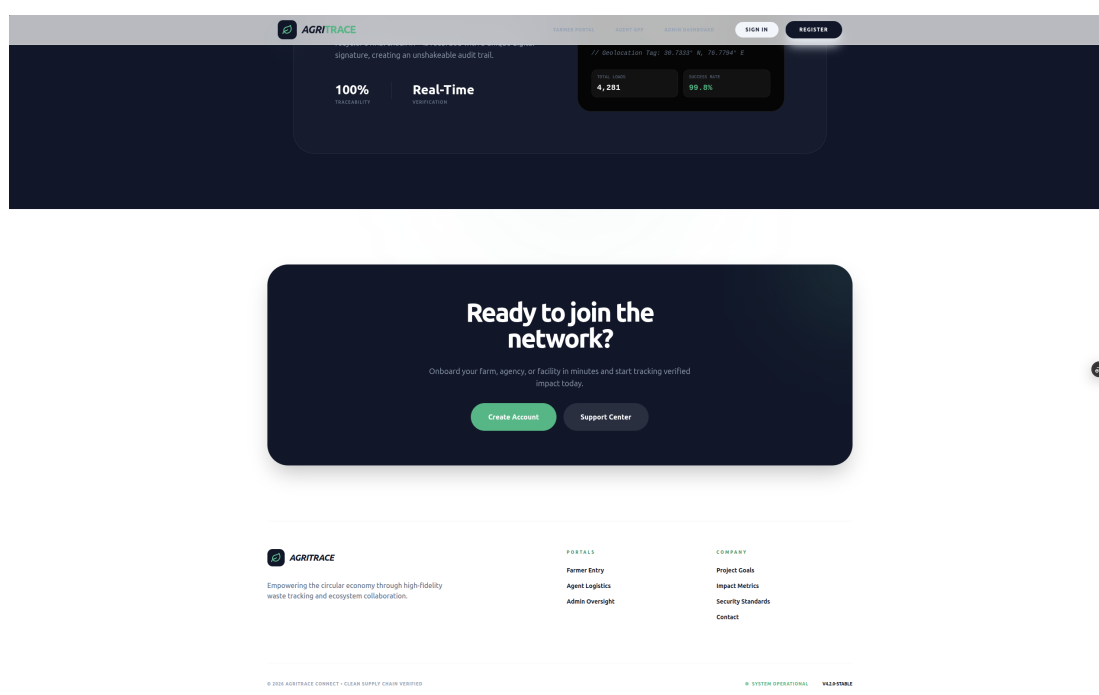


Figure 9.9: Contact Us Section

9.2 Landing Page

The AgriTrace landing page serves as the primary entry point for new and returning users. The page is designed with a modern, responsive layout that communicates the platform's value proposition effectively.

- **Hero Section:** The top section features a bold headline (“Transform Agricultural Waste

into Value”) with a gradient text effect, a descriptive subtitle explaining the platform’s purpose, and two call-to-action buttons: “Get Started” (primary, links to signup) and “Learn More” (secondary, scrolls to features section). The background uses a subtle gradient from green to earth tones.

- **Feature Bento Grid:** Below the hero, a CSS Grid-based bento layout showcases six core platform features:

1. *Waste Tracking* – End-to-end visibility from farm to recycling facility
2. *Collection Management* – Coordinated pickup logistics
3. *Data Analytics* – Real-time charts and statistics
4. *Mobile Responsive* – Optimized for all device sizes
5. *Smart Verification* – Photo-based waste verification
6. *Complete History* – Full audit trail of all transactions

Each feature card includes a Lucide icon, title, and description with hover animation effects.

- **Statistics Section:** An animated counter section displays key platform metrics (e.g., total waste diverted, active users, carbon offsets generated) with smooth number counting animations.
- **Footer:** Contains navigation links, contact information, and copyright notice.

9.3 Authentication Module

9.3.1 Login Page

The login page provides a split-screen layout:

- **Left Panel:** Contains the login form with:
 - AgriTrace logo and welcome message
 - Email input field with icon prefix
 - Password input field with show/hide toggle
 - “Forgot Password?” link that triggers Firebase’s password reset email flow
 - “Sign In” button with loading spinner during authentication
 - “Don’t have an account? Sign up” link for new users
- **Right Panel:** A decorative section featuring:
 - Rotating testimonial cards with fade-in/fade-out animations

- Each testimonial displays a user quote, name, role, and avatar
- Three testimonials cycle every 5 seconds
- **Error Handling:** Firebase error codes are mapped to user-friendly messages:
 - “auth/user-not-found” → “No account found with this email”
 - “auth/wrong-password” → “Incorrect password”
 - “auth/too-many-requests” → “Too many attempts. Try again later”

9.3.2 Signup Page

The registration page allows new users to create accounts:

- Name, email, and password fields with real-time validation
- Role selection dropdown (Farmer, Agent) – Admin role is assigned manually
- Password strength indicator showing minimum requirements
- Terms and conditions checkbox
- Upon successful registration, users are redirected to a role-specific onboarding screen

9.4 Farmer Dashboard

The Farmer Dashboard is the primary interface for agricultural waste producers. It is organized into several functional sections:

9.4.1 Statistics Panel

Four StatCard components display key metrics:

1. **Total Listings:** Count of all listings ever created by the farmer
2. **Active Listings:** Count of listings in OPEN or ASSIGNED status
3. **Delivered:** Count of listings that have been delivered to recyclers
4. **Recycled:** Count of listings marked as fully recycled by admin

Each card features a colored icon, metric value (with animated counter), and a brief description. The color coding provides visual differentiation: blue for total, green for active, amber for delivered, and purple for recycled.

9.4.2 Create New Listing

A dialog-based form allows farmers to create waste listings:

- **Title:** Free-text input for listing name (e.g., “Rice husk from paddy field”)
- **Category:** Dropdown selector with options: Rice Husk, Rice Residue, Wheat Stubble, Sugarcane Bagasse, Cotton Stalks, Corn Stover, Paddy Straw, Coconut Shell, Groundnut Shell, Mustard Stalks, Other
- **Quantity:** Numeric input in metric tonnes
- **Price:** Price per unit in Indian Rupees (INR)
- **Description:** Multi-line textarea for additional details
- **Location:** GPS-based `LocationPicker` component that captures coordinates and displays a reverse-geocoded address
- **Photos:** `PhotoUploader` component allowing multiple image uploads (up to 5 images) via Cloudinary

9.4.3 Listing Management

Two tabs organize the farmer’s listings:

- **Active Listings:** Shows all listings not in CANCELLED or RECYCLED status. Each card displays the title, category badge, quantity, price, status badge (color-coded), assigned agent (if applicable), and creation date. A “Cancel” button is available for OPEN listings.
- **Completed Listings:** Shows RECYCLED and CANCELLED listings with delivery/-cancellation timestamps.

9.4.4 CSV Export

A “Download CSV” button generates a comma-separated file containing all listing data (title, category, quantity, price, status, dates) for offline record-keeping and accounting purposes.

9.5 Agent Dashboard

The Agent Dashboard provides collection agents with task management capabilities:

9.5.1 Statistics Panel

Four `StatCard` components display agent-specific metrics:

1. **Available Tasks:** Count of OPEN listings available for claiming

2. **Assigned Tasks:** Count of listings assigned to this agent
3. **In Transit:** Count of listings currently being transported
4. **Delivered:** Count of successfully delivered listings

9.5.2 Available Tasks

A scrollable list of OPEN listings that the agent can claim. Each task card shows:

- Listing title and category
- Quantity and offered price
- Farmer's location (address text and distance from agent's current location)
- "Claim" button that atomically assigns the listing to the agent

9.5.3 My Tasks

Shows all listings assigned to the current agent, organized by status:

- **ASSIGNED:** "Start Pickup" button transitions the listing to IN_TRANSIT
- **IN_TRANSIT:** "Pay & Deliver" button triggers the integrated Razorpay payment flow. Upon successful verification of the payment signature, the system atomically updates the listing status to DELIVERED, marks the payment as PAID, and links the transaction's unique order ID to the listing document.
- Action buttons use confirmation dialogs to prevent accidental status changes

9.6 Admin Dashboard

The Admin Dashboard provides platform-wide oversight and management capabilities:

9.6.1 Platform Statistics

Comprehensive statistics cards display:

- Total registered users (broken down by role: farmers, agents)
- Total listings (broken down by status)
- Total waste diverted (in metric tonnes)
- Total carbon offsets generated (in kg CO₂)
- Total transaction value (in INR)

9.6.2 User Management

A tabular view of all registered users with columns for name, email, role, registration date, and verification status. Admin actions include:

- Edit user role (change between farmer, agent, admin)
- View user activity history
- Filter users by role using tab-based navigation

9.6.3 Listing Oversight

A complete view of all platform listings across all users. Admin-specific actions include:

- Mark DELIVERED listings as “Recycled” to complete the waste lifecycle
- View listing details including photos, location, and assignment history
- Filter listings by status using dropdown selectors

9.6.4 Analytics Dashboard

Interactive charts built with the Recharts library provide visual insights:

- **Waste by Type:** Pie chart showing distribution of waste categories across all listings
- **Collection Over Time:** Line/bar chart showing listing creation and completion trends over time
- **Status Distribution:** Bar chart comparing listings across all status categories
- Charts are responsive and include tooltips for detailed data inspection

9.7 Payment Processing Flow

The Razorpay integration provides a complete payment workflow:

1. The `PaymentButton` component appears on DELIVERED listings
2. Clicking the button triggers a server-side API call to create a Razorpay order
3. The Razorpay checkout modal opens with pre-filled amount and merchant details
4. The modal supports multiple payment methods: UPI (scan QR or enter UPI ID), debit/-credit cards, net banking, and wallets
5. Upon successful payment, a success toast is displayed and the listing’s payment status is updated
6. Payment confirmations are stored in the orders collection for audit purposes

9.8 Notification System

The notification system provides real-time updates through an in-app notification center:

- A bell icon in the navigation header displays the unread notification count as a badge
- Clicking the bell opens a dropdown panel showing recent notifications
- Notification types include:
 - *Info*: General platform updates and announcements
 - *Success*: Successful operations (listing created, payment received)
 - *Warning*: Action required (new listing claimed, pickup started)
 - *Error*: Failed operations or system issues
- Each notification can be marked as read individually
- Notifications are sorted by creation time (newest first)

9.9 Carbon Credits Display

The carbon credits section provides environmental impact quantification:

- A dedicated panel on the farmer dashboard shows total carbon offsets earned
- Individual listing contributions are itemized with waste type, quantity, and calculated CO₂ savings
- A cumulative progress bar shows progress toward environmental milestones
- Carbon offset values are calculated using the scientifically established factors defined in the `CARBON_FACTORS` constant

9.10 Responsive Design

The application adapts its layout across three primary breakpoints:

- **Desktop (1024px+)**: Full-width layout with sidebar navigation, multi-column card grids, and side-by-side form layouts
- **Tablet (768px–1023px)**: Reduced column count (2-column grids), collapsible navigation, and stacked form fields
- **Mobile (below 768px)**: Single-column layout, hamburger menu navigation, full-width cards, and bottom sheet dialogs instead of modal dialogs

Tailwind CSS responsive modifiers (`sm:`, `md:`, `lg:`, `xl:`) are used consistently throughout the codebase to implement these breakpoints.

The figures above are referenced in the final output chapter and can be updated anytime by replacing image files in the report image folder.